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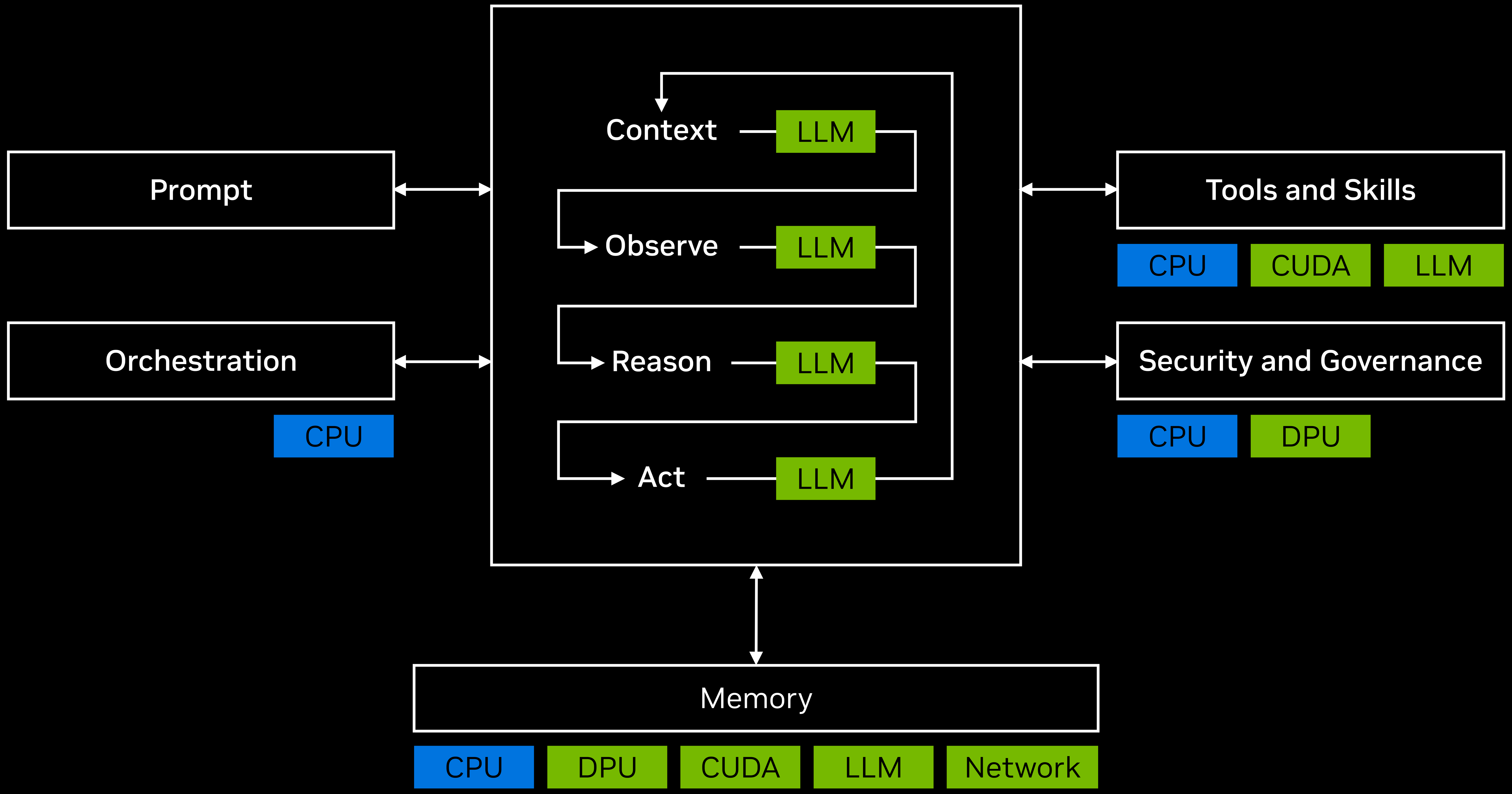
NVIDIA Spectrum-X Ethernet Multiplane Network Architecture

Gilad Shainer

Hot Chips 2026

Agentic AI Is the Most Complex Computing Workload in History



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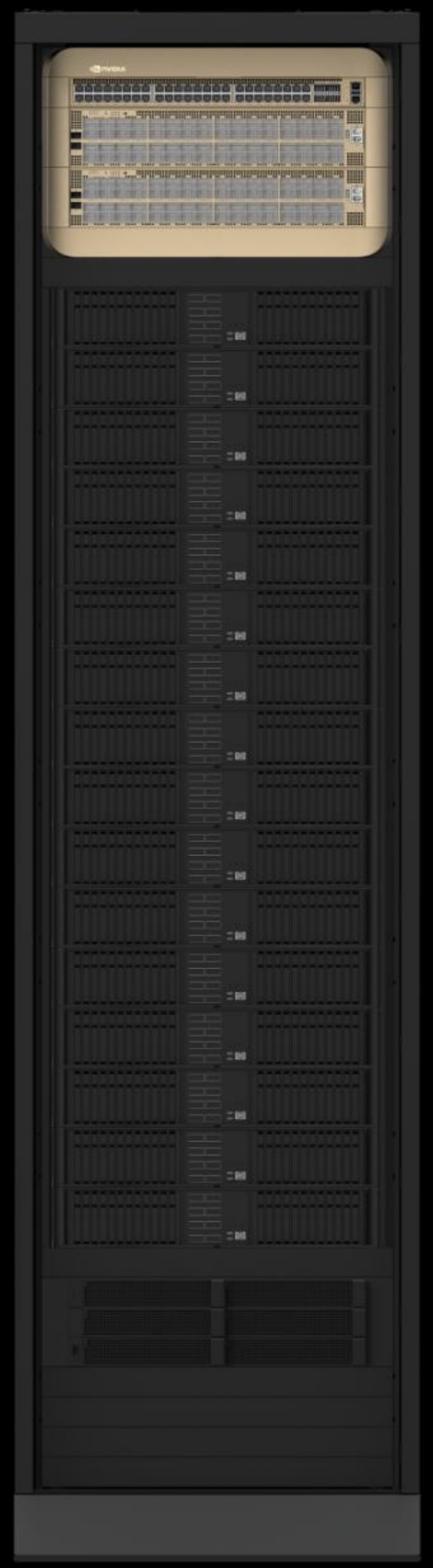
NVIDIA Vera Rubin Is a Full-Stack AI Factory Platform

Vera Rubin
NVL72

Groq 3
LPX

Vera CPU
Rack

Vera BlueField-4
STX Storage



Spectrum-6 SPX
Networking

AI Factories Need Five Purpose-Built Networks
Not One General-Purpose Fabric

NVIDIA NVLink Scale-up Networking Fabric for AI Factories

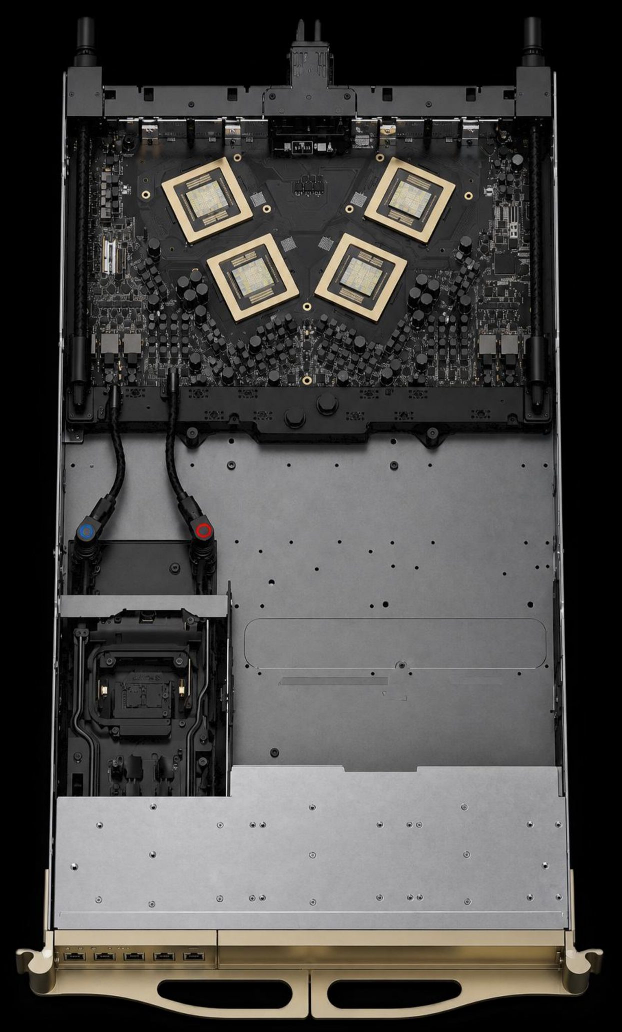
Purpose-built scale-up fabric delivers leading performance and intelligent resiliency

NVL72 Rack

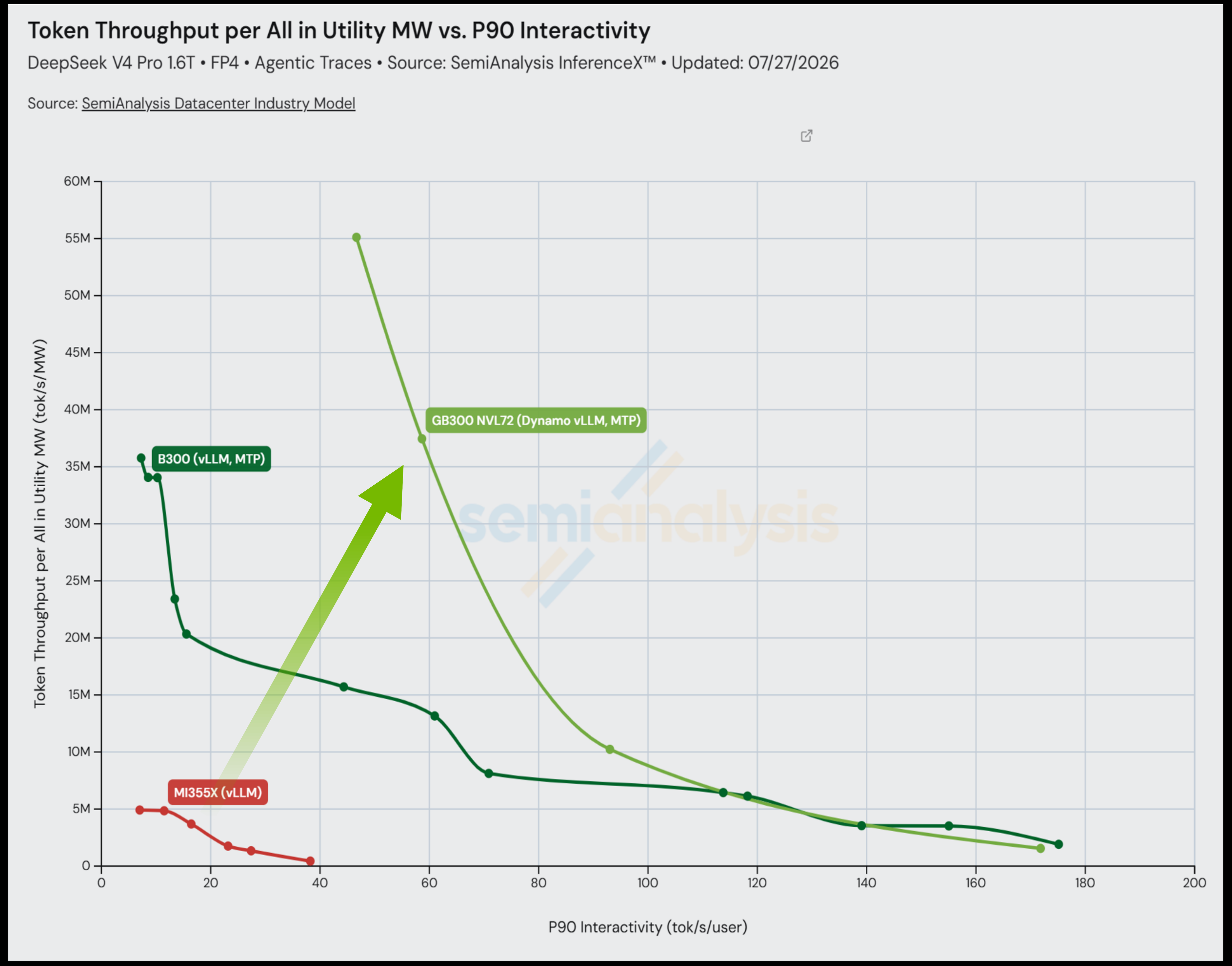
72 GPU scale-up domain

NVLink Spine

NVLink Switch Tray



NVLink NVL72 Enables Leading tokens/MW



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The Different Ethernet Architectures

AI Factories require purpose-built infrastructure

Enterprise

Enterprise,
feature-rich DC

Hyperscale Spine

Hyperscale DC,
cloud spine

AI Factories

High-performance,
distributed computing

Service Provider

Service provider core,
carrier, DCI

Spectrum-X:
The Ethernet Architecture for Giga-Scale AI Factories

Scaling AI factories of hundreds of thousands of GPUs

The only purpose-built scale-out Ethernet for AI workloads

Adaptive routing, congestion control, jitter-free communications

End-to-end with 102.4T switch systems and 1.6T SuperNICs

Large ecosystem adoption and open operating systems

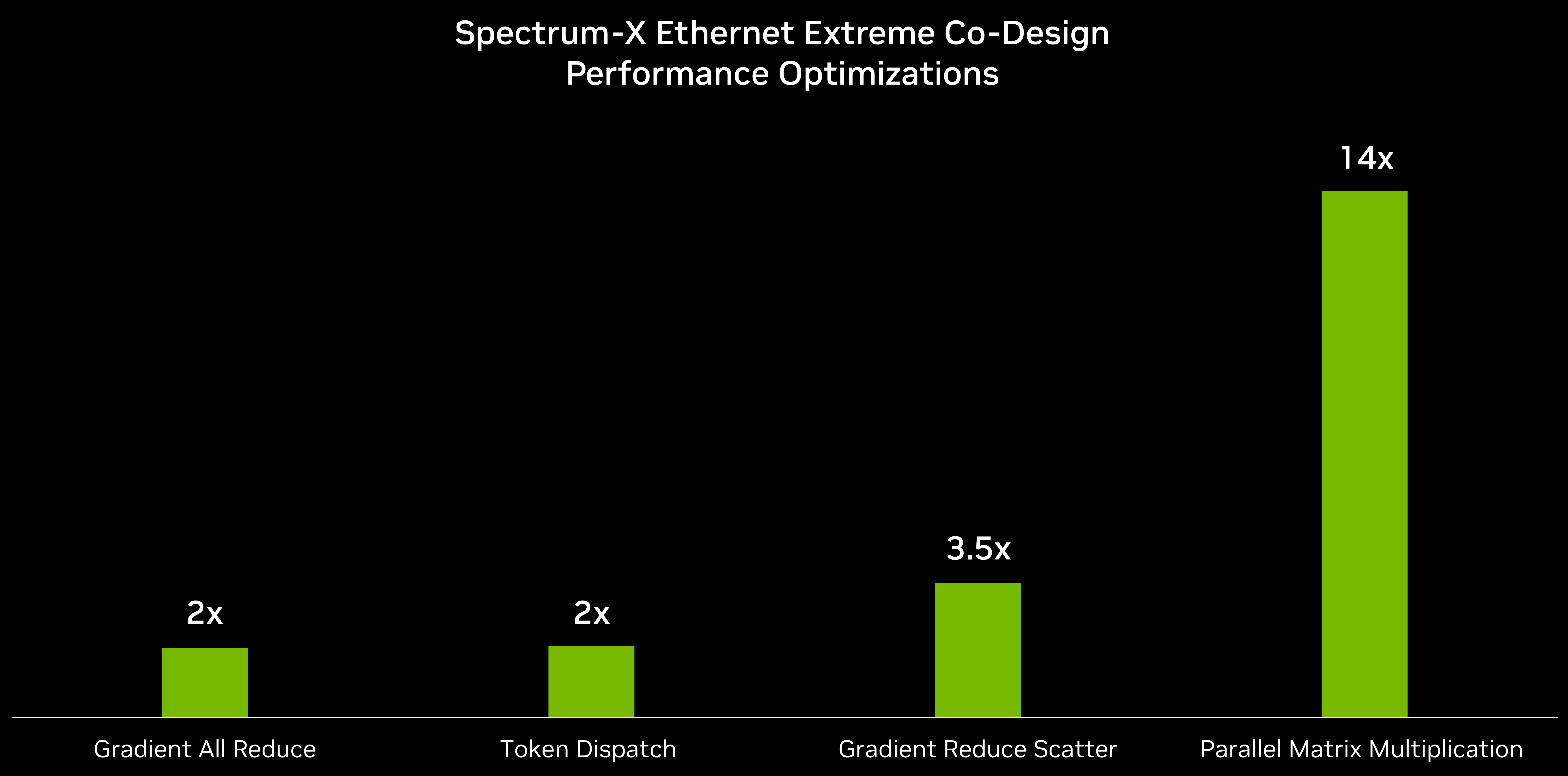
1.6x	2.2x	1.3x
Higher RDMA bandwidth	Multi-tenancy bandwidth	Lower jitter

1.9x Higher Training Performance in Multi-Tenant AI Factory

Spectrum-X Ethernet low-jitter communications, multi-tenant noise isolation
Higher performance with deterministic results

Up to 14x Higher NCCL Performance

The main NCCL operations for Nemotron Ultra pre-training



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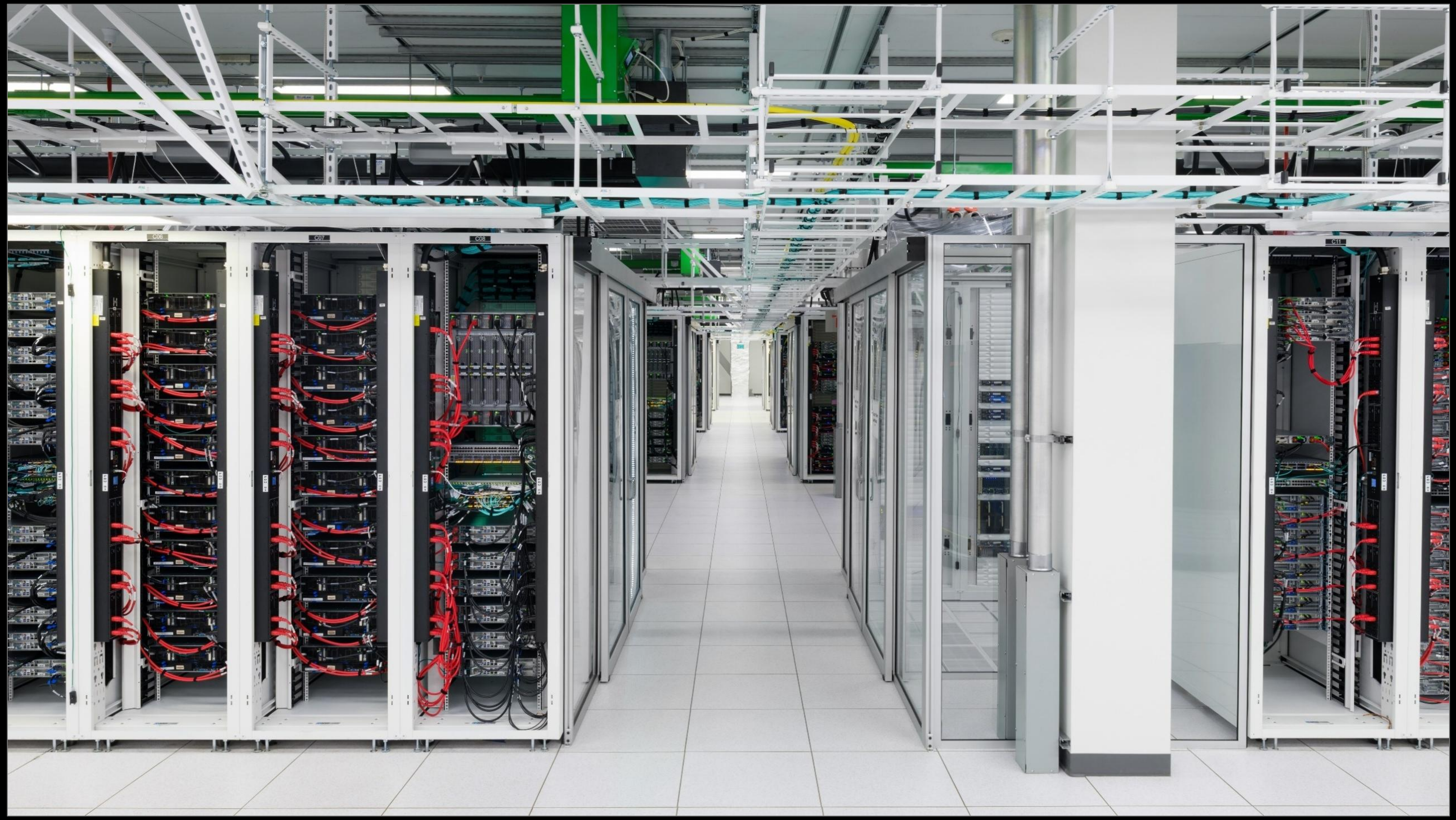
Scale-Out AI Depends on Optics

Equal to 10% of compute power



Traditional Cloud Data Center

AI Factory

Spectrum-X Ethernet Photonics Increases AI Capacity and Productivity

Scaling AI factories of hundreds of thousands of GPUs

CPO chip with micro ring modulator technology in production

3D-stacked silicon photonics engine with TSMC Coupe process

High-power, high-efficiency lasers

Detachable fiber connectors

4x	5x	10x
Less lasers	Lower power	Lower MTBI

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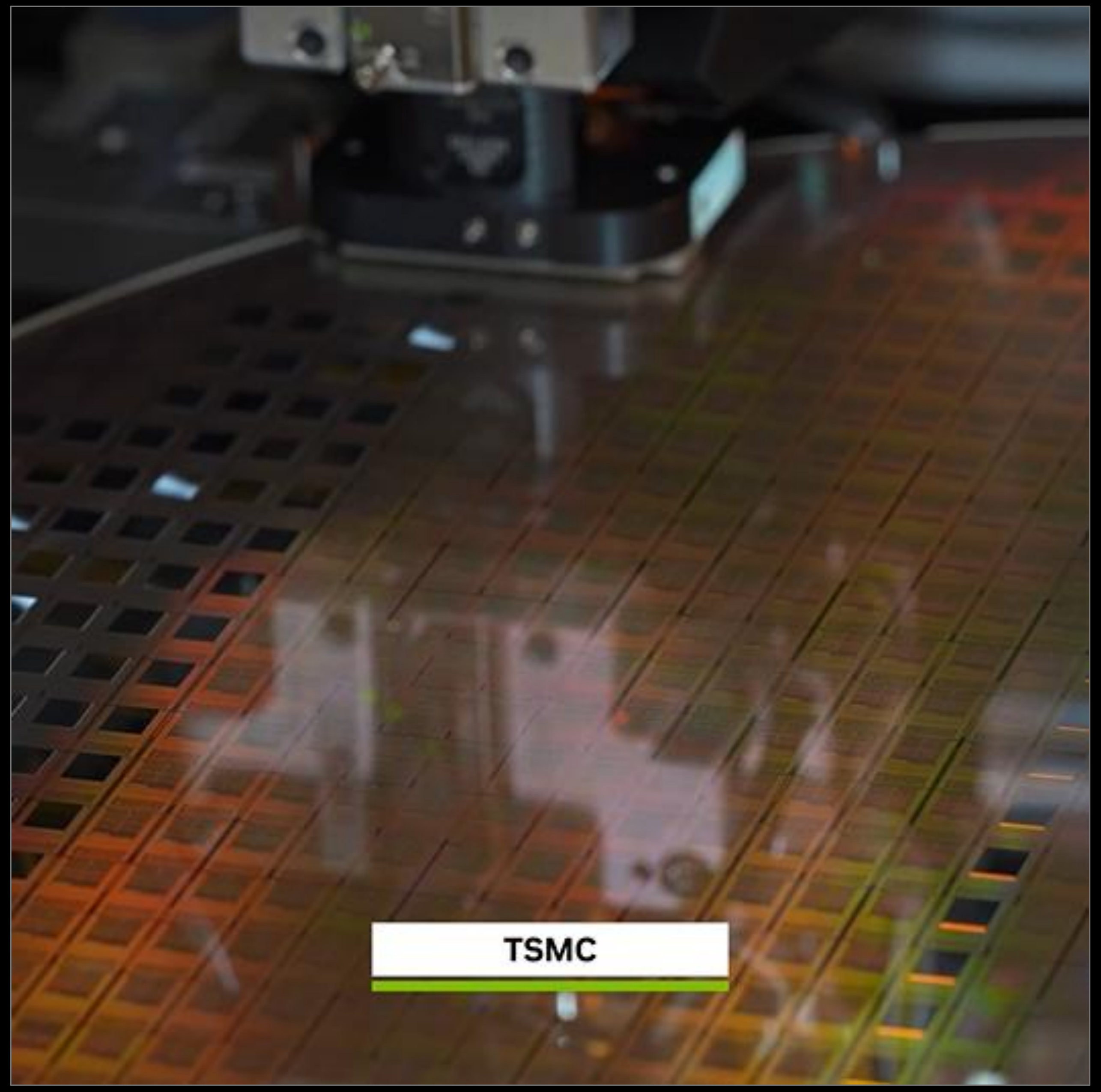
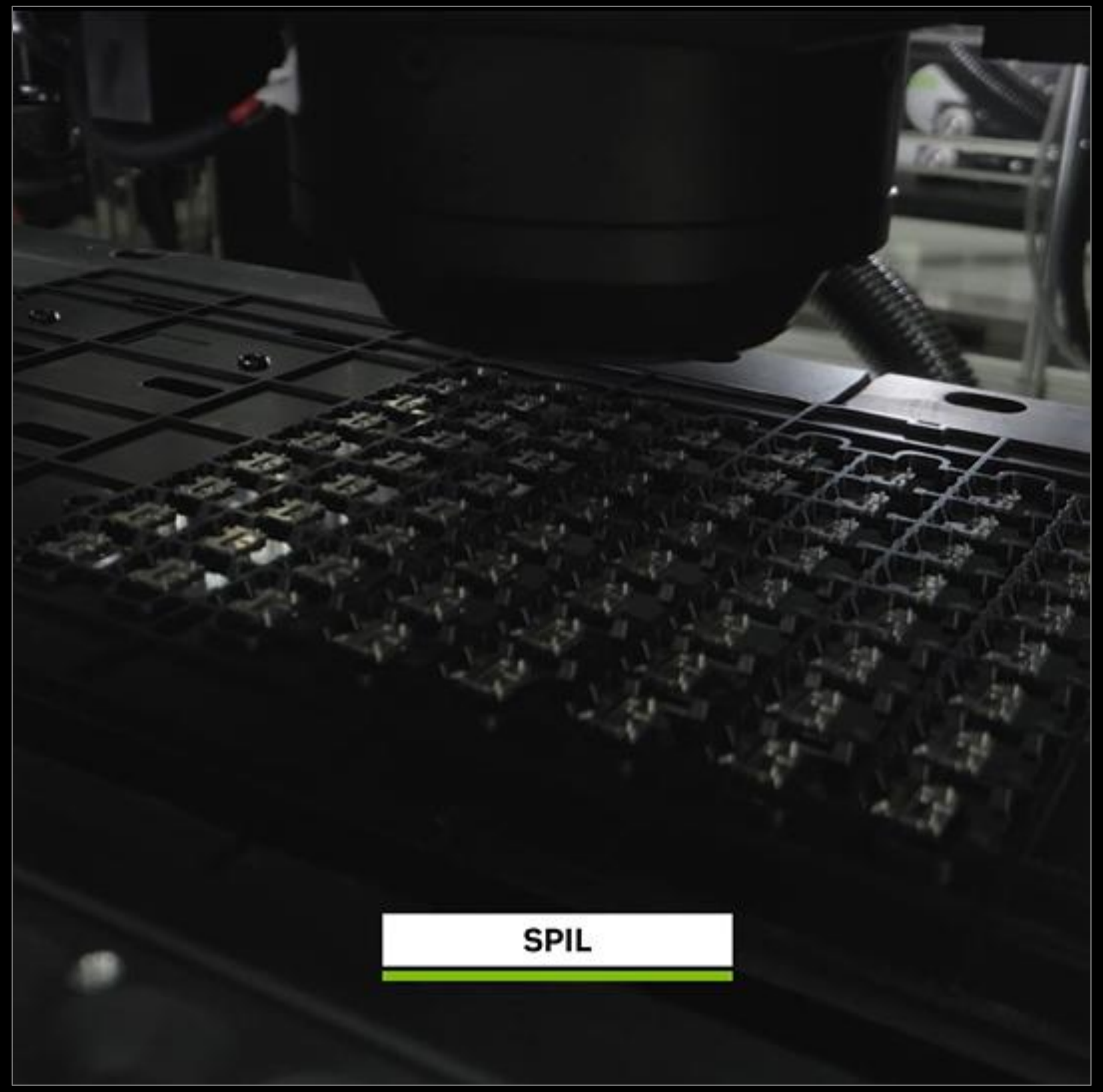
NVIDIA Photonics in Production





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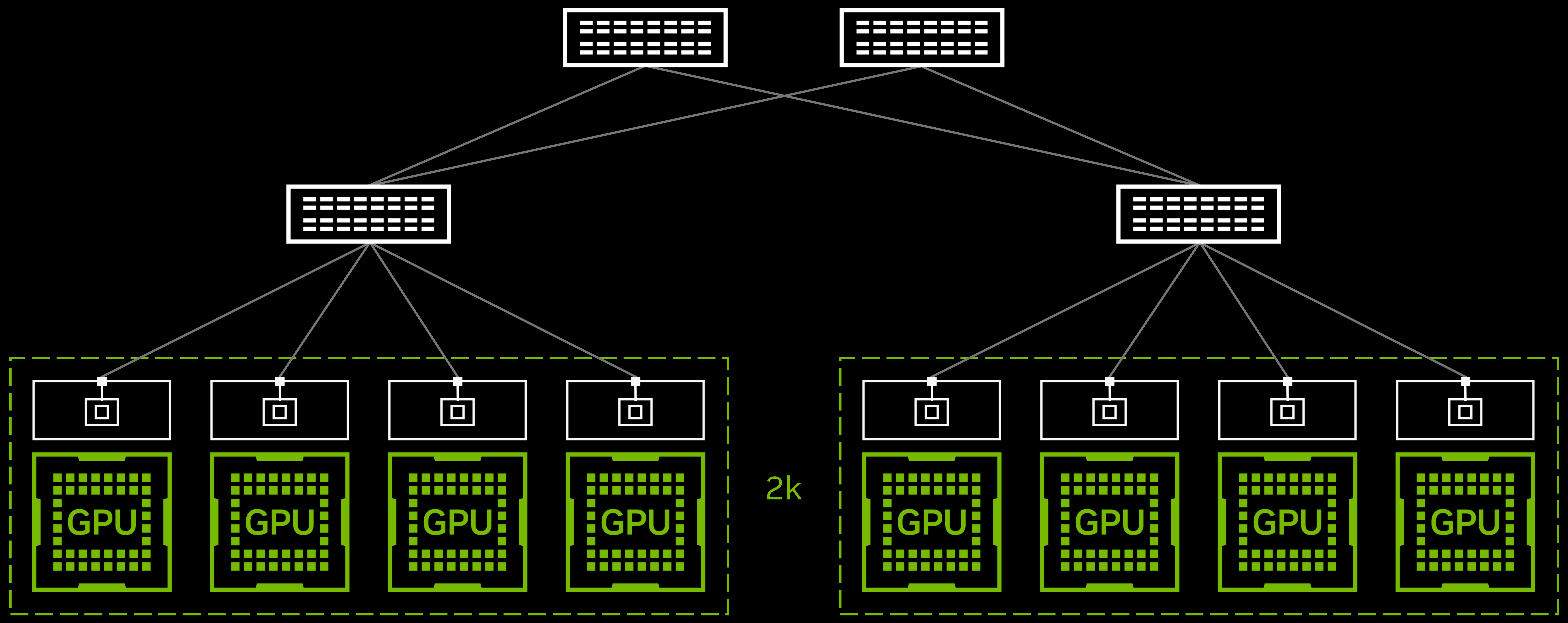
Traditional Hyperscale Cloud Top-of-Rack Switch Topology



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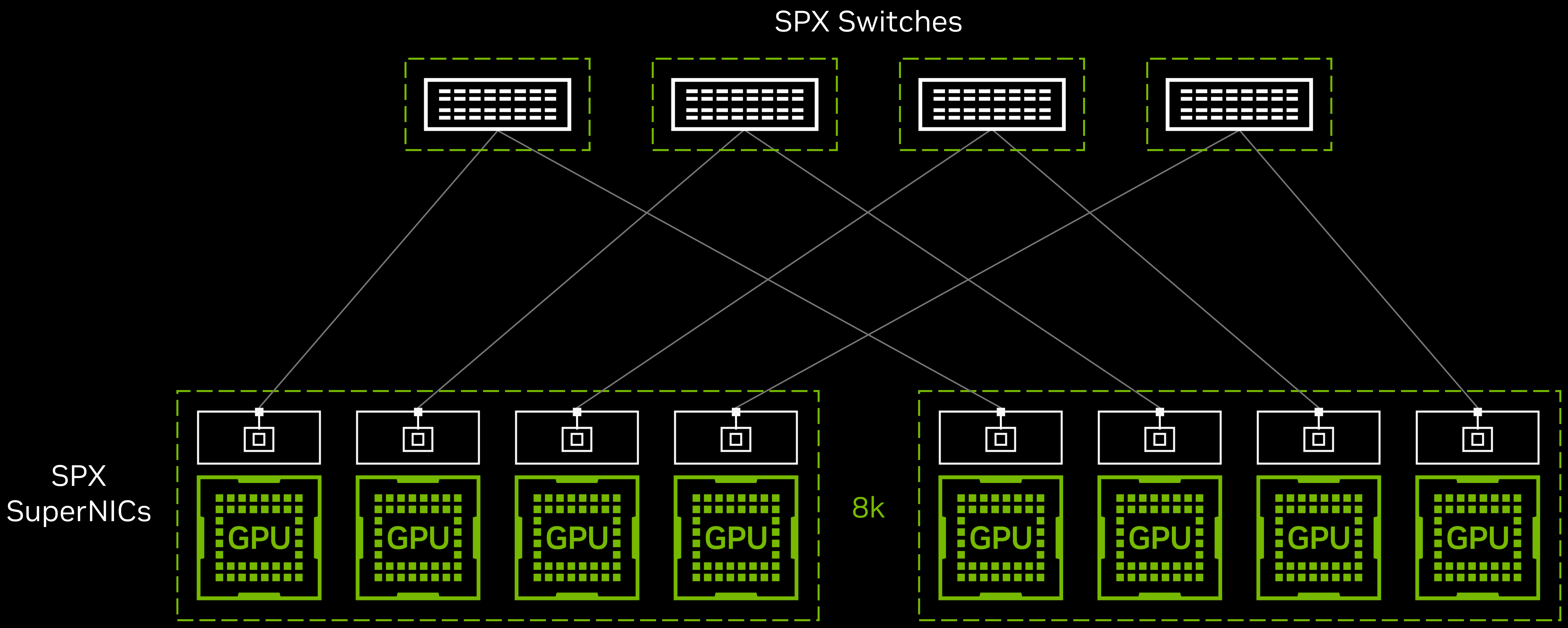
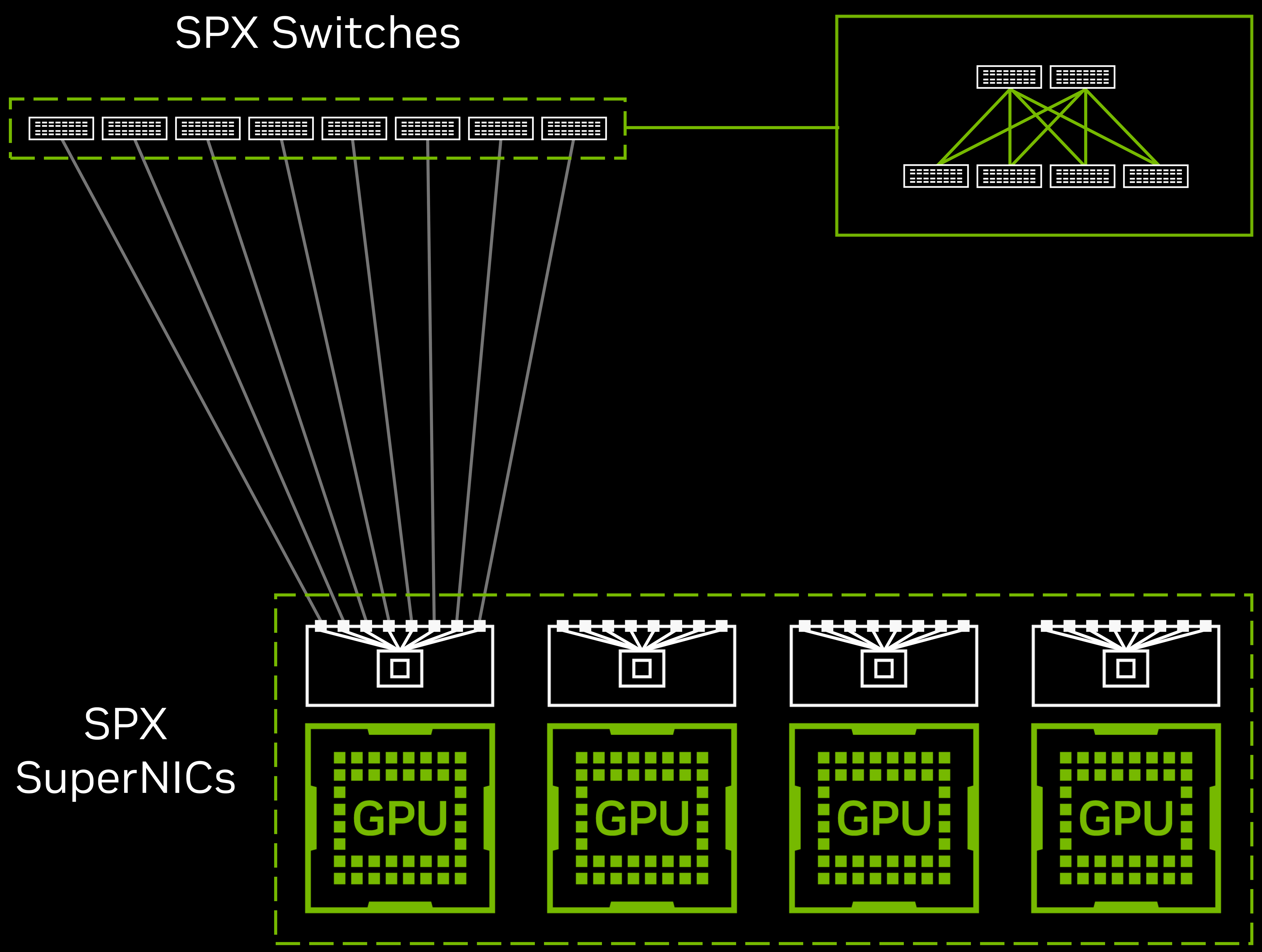
AI Factory With Spectrum-X Ethernet Multi-Rail Topology



8k Rubin GPUs, at 1.6T scale-out bandwidth per GPU, with 100T switches supporting 64 ports of 1.6T, 4 rails

Introducing Spectrum-X Ethernet Multiplane Topology



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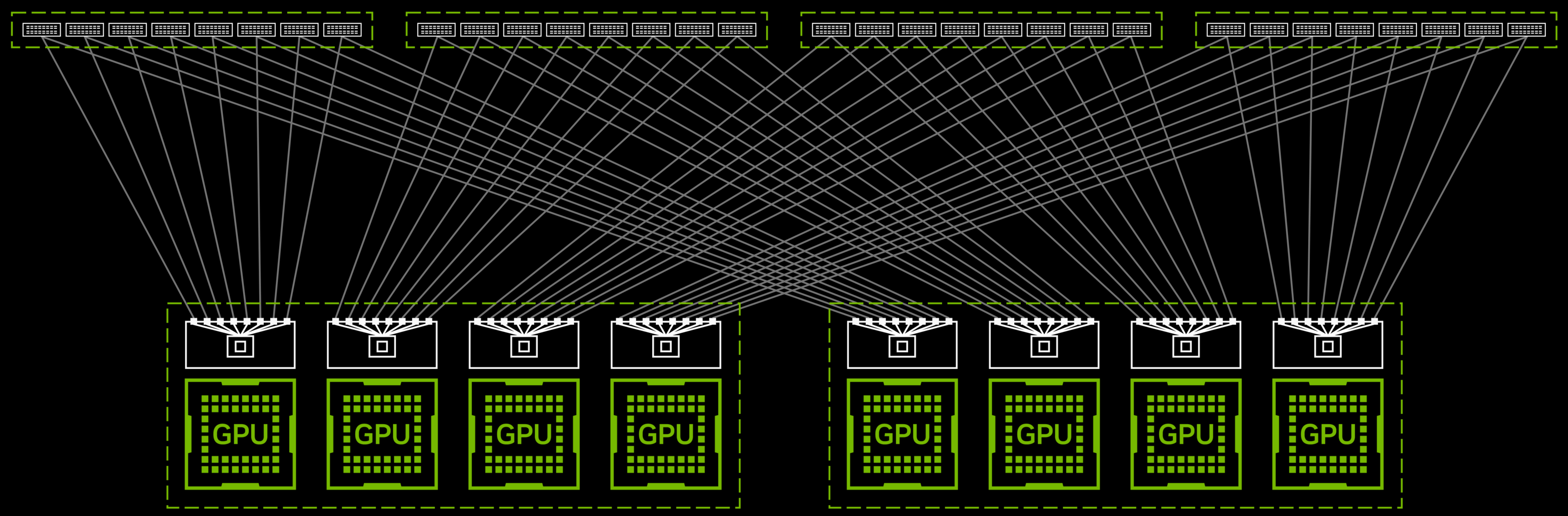
Spectrum-X Ethernet Multiplane Extends AI-Factory Scale by 64X

Hardware accelerated Spectrum-X Ethernet multiplane topology
Lower latency, higher resiliency, hardware-based global adaptive routing, and congestion control

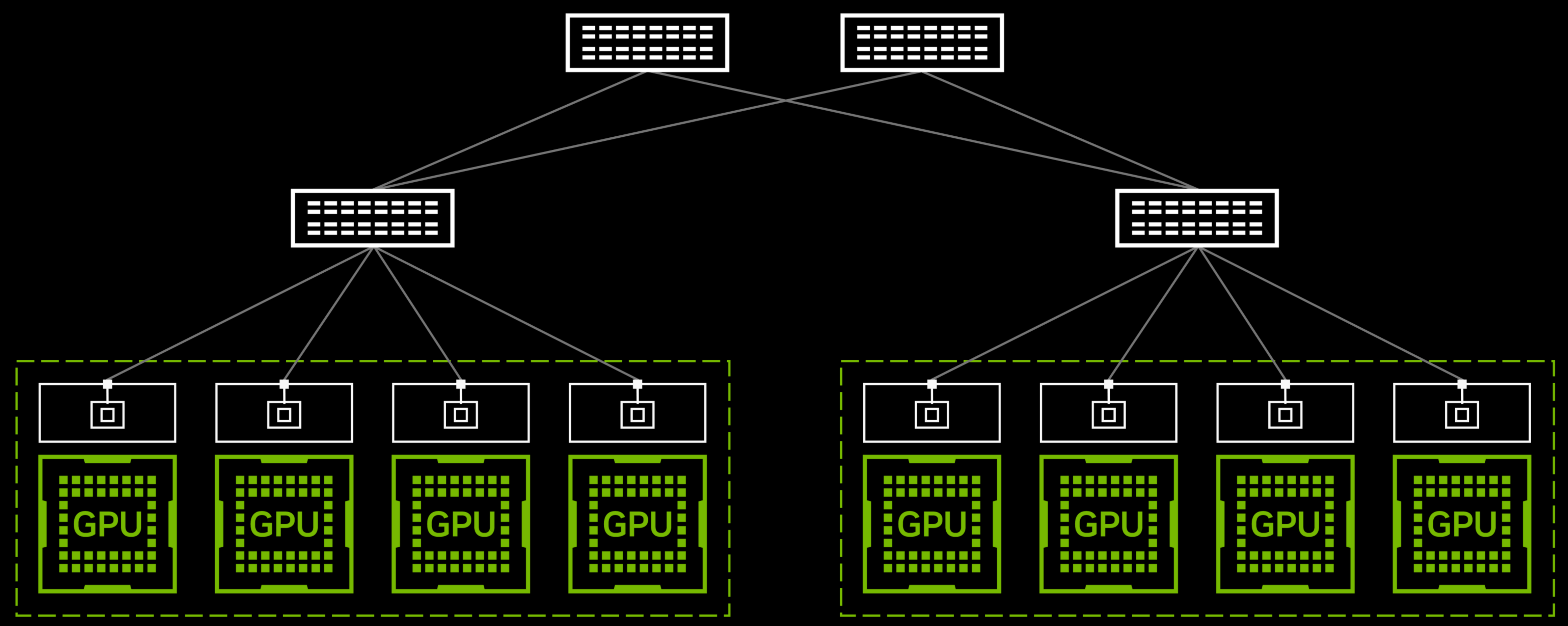
512k Rubin GPUs, at 1.6T scale-out bandwidth per GPU, with 100T switches supporting 512 ports of 200G, 8 planes, 4 rails

Spectrum-X Ethernet Multiplane Saves 1.7x Scale-Out Switches

Power, rack-space, cost



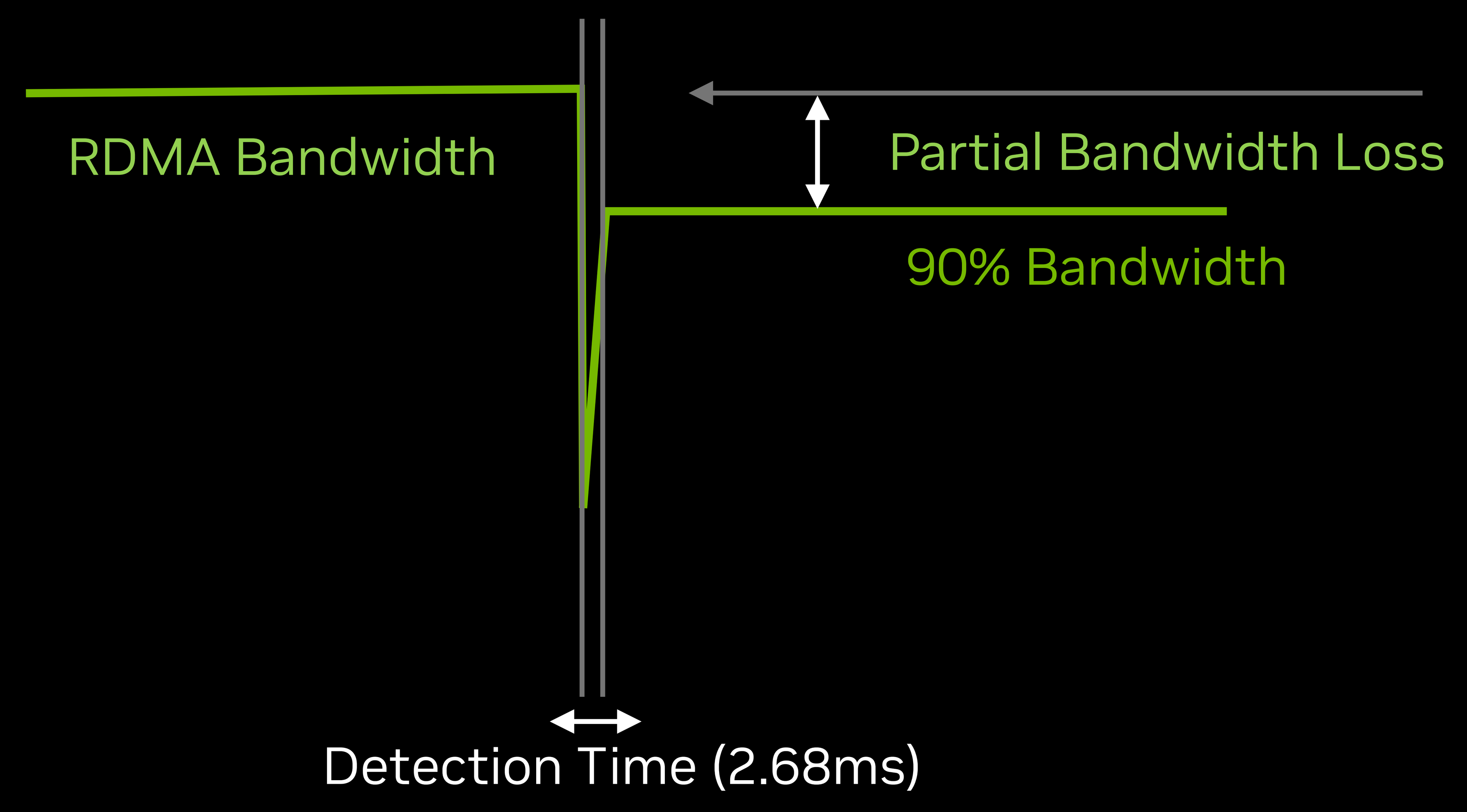
Spectrum-X Ethernet Multi-Plane Multi-Rail Topology



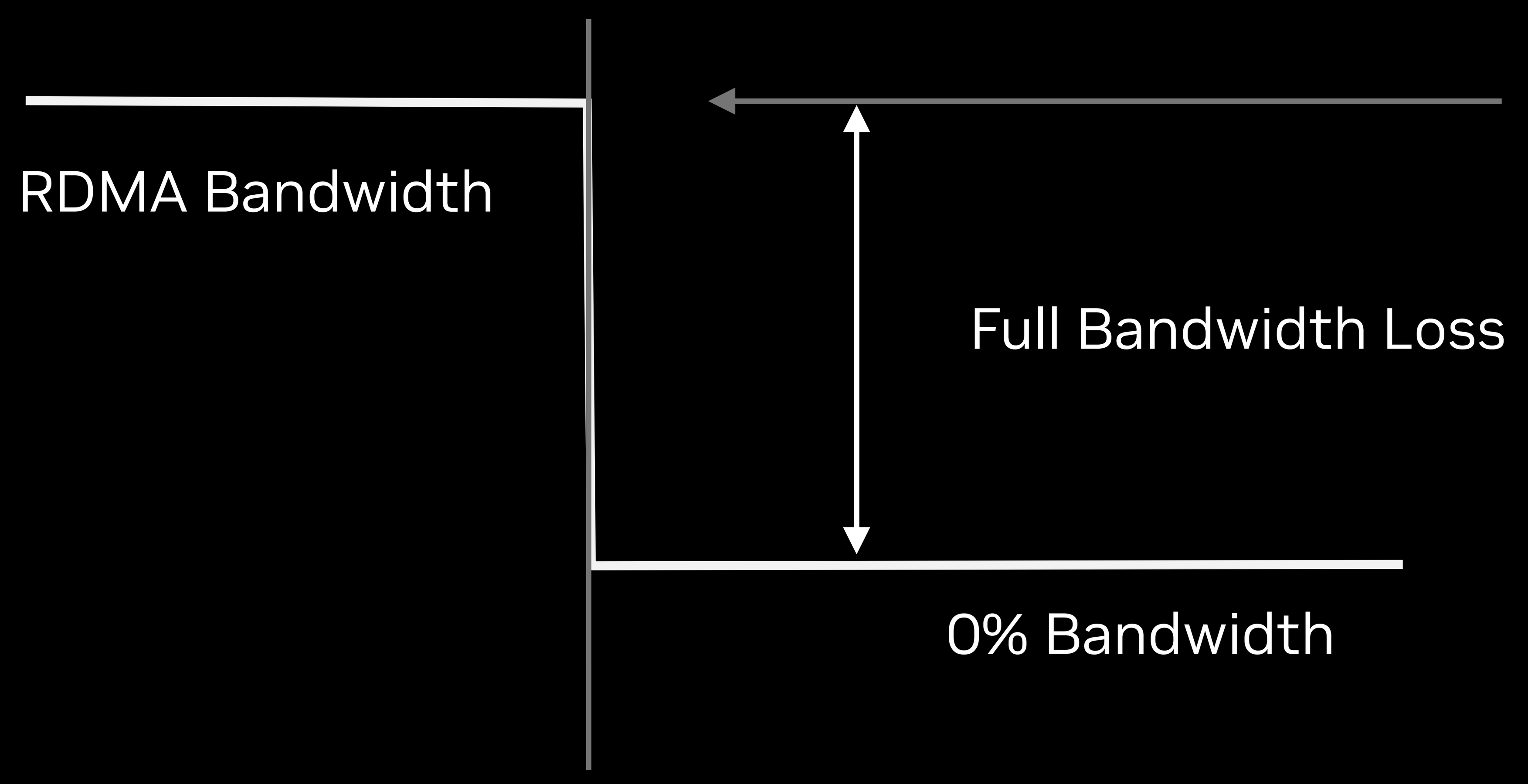
Traditional Multi-Tier Single Rail Switch Topology

Spectrum-X Ethernet Multiplane Maintains 90% Bandwidth

Spectrum-X Ethernet
Multiplane Topology



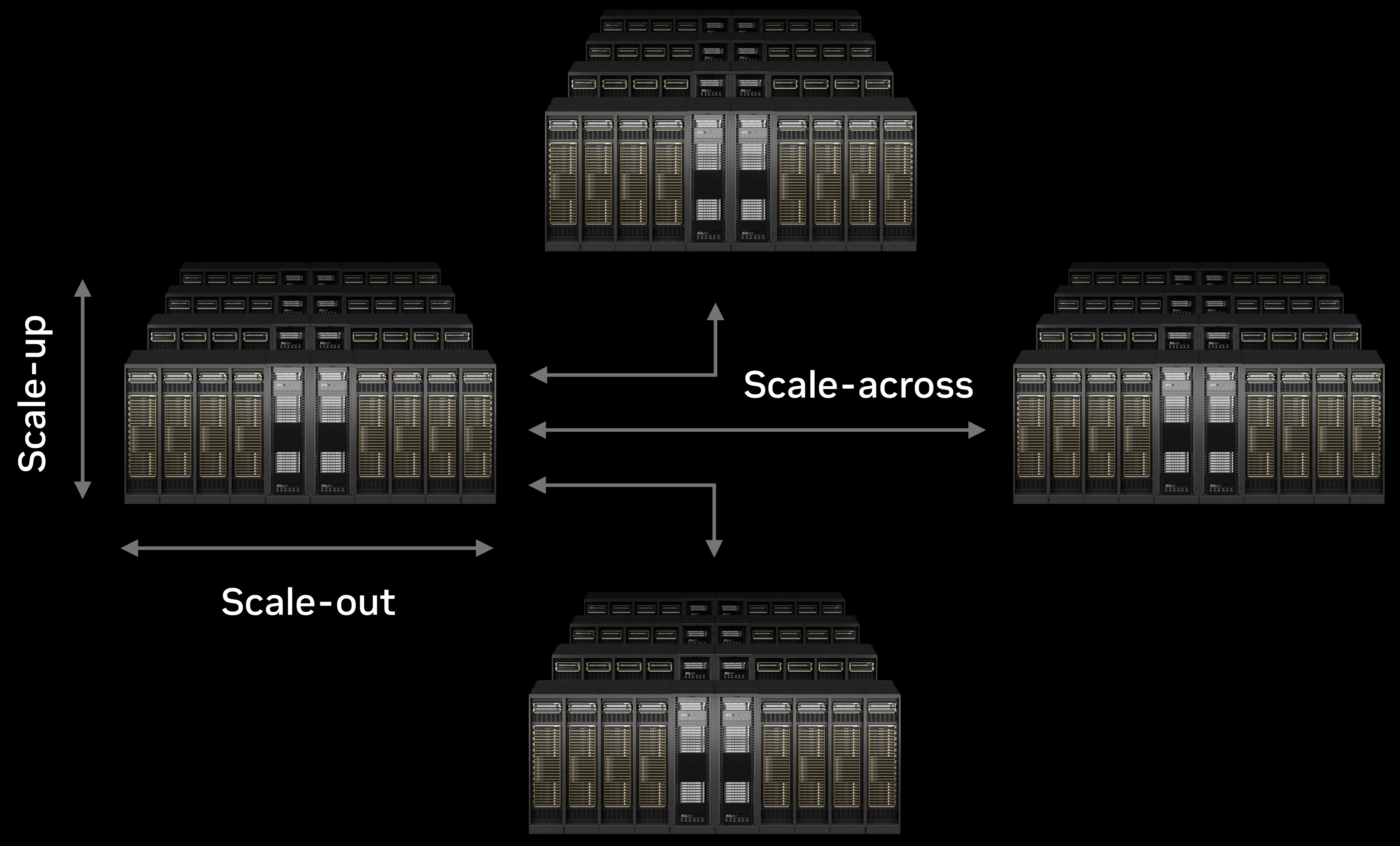
Traditional Multi-Tier
Switch Topology



Spectrum-X Ethernet Multiplane Delivers 1.6X Higher AI-Factory Goodput

Spectrum-XGS Connects AI Factories Into Giga-Scale Infrastructure

Doubled scale-across performance for multi-site distributed AI operations



1.9x

Multi-site performance

Spectrum-X Ethernet Switch

800Gb/s per port | 102.4Tb/s

ConnectX-9 SuperNIC

800Gb/s networking

- AI Factories with open NOS support
- Distance-aware load balancing and congestion control
- Unlimited scale across cities, states, continents
- Minimize cross data-center latency

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NVLink Fusion Connects XPUs to the NVIDIA AI Platform

Deploy XPUs with faster time-to-market, leading NVLink scale-up performance, and unified MGX rack architecture and ecosystem

Compute Tray x18
Cableless, Hoseless, Modular Design
45C Inlet Temp

NVIDIA NVLink Switch Tray x9
3.6 TB/s per XPU All-to-All Bandwidth
3X Lower Latency
Built-In Factory Resiliency

NVIDIA NVLink Spine x1
Connects 72 XPUs in a Single Domain

NVIDIA MGX Rack
Third-Generation Design
100% Liquid Cooled

NVIDIA DSX AI Factory Platform

Extreme co-design at infrastructure scale

DSX OS

Power Optimization

Infrastructure Software

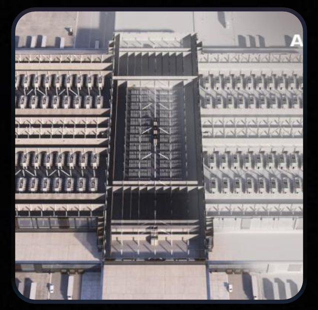
Platform Software

Libraries and Frameworks

DSX Sim

DSX MaxLPS

DSX Flex



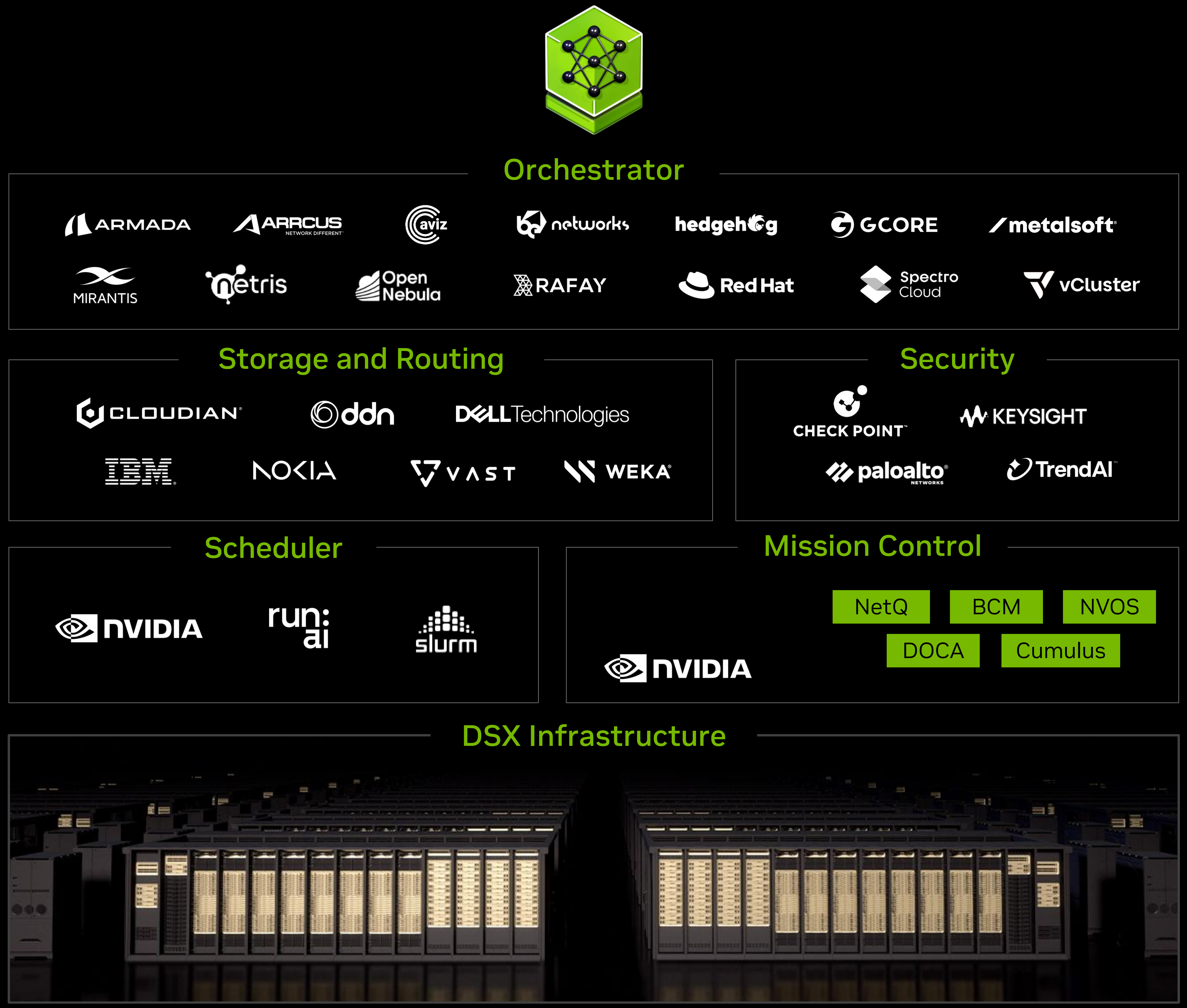
Reference Designs

AI Factory Requires Simulation and Validation at Scale

DSX Air Compresses AI-Factory Deployment From Months to Days

AI Factory design and validation platform

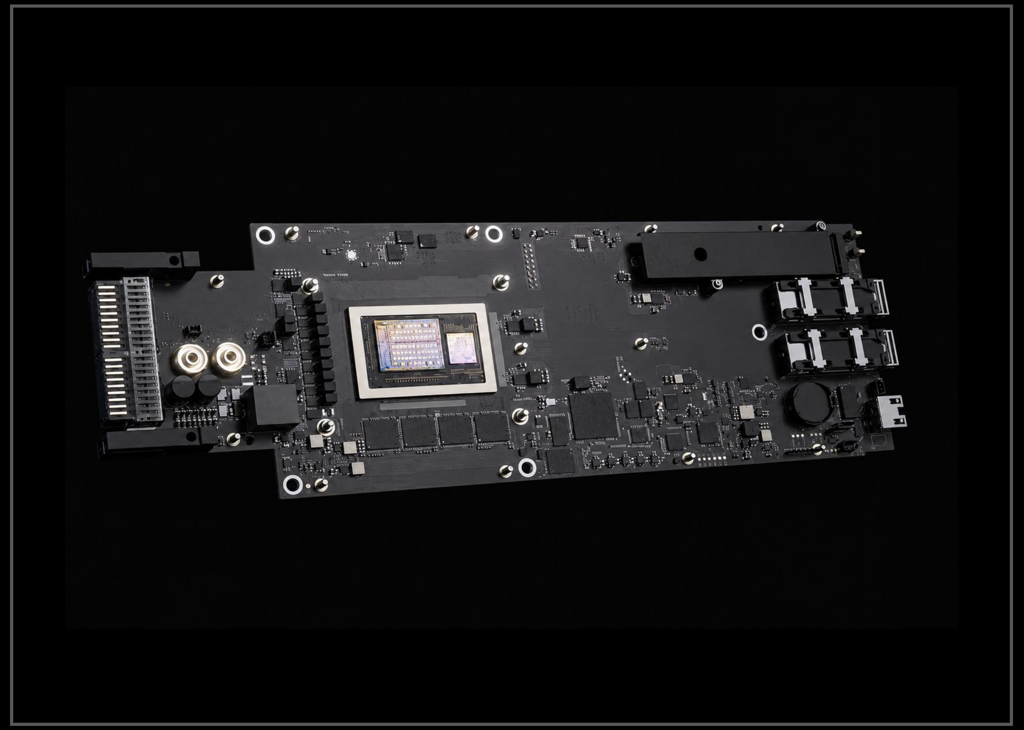
6 Months to 1 Week	3 Months to 1 Week	2 Weeks to 1 Day
Infrastructure validation	Software bring-up	Deployment time



The Five Networking Infrastructures of NVIDIA AI Factory

The only purpose-built networking that scales every dimension for the most efficient and resilient AI factories

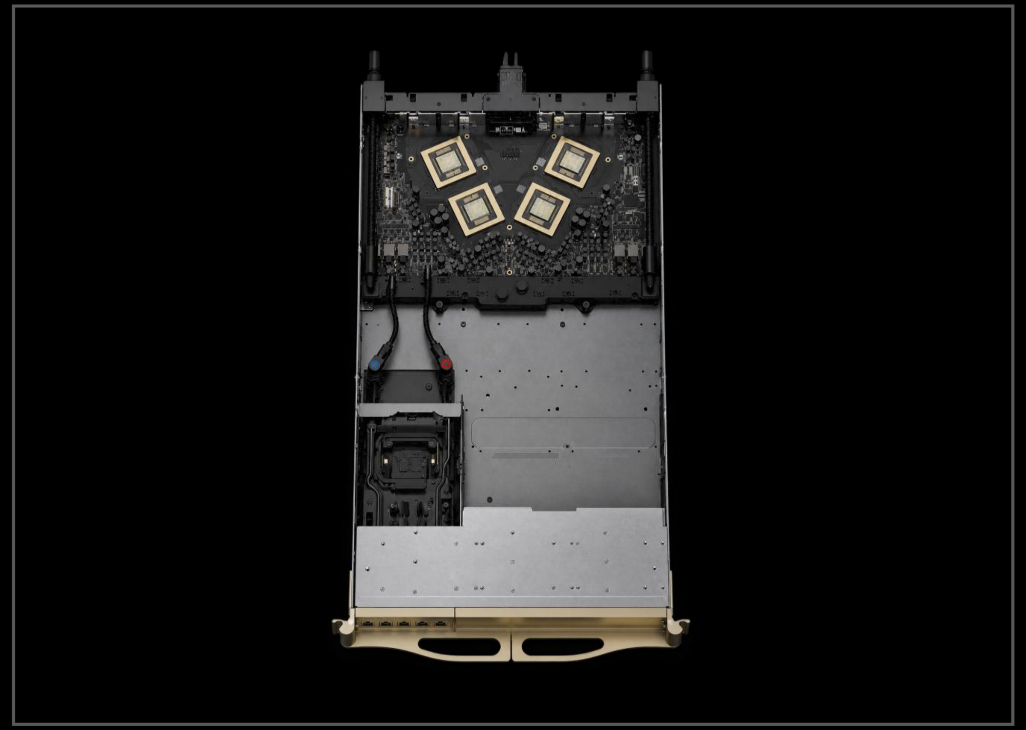
Scale-In
BlueField-4 DPU



18X
AI Factory Services bandwidth

2X
Storage Acceleration

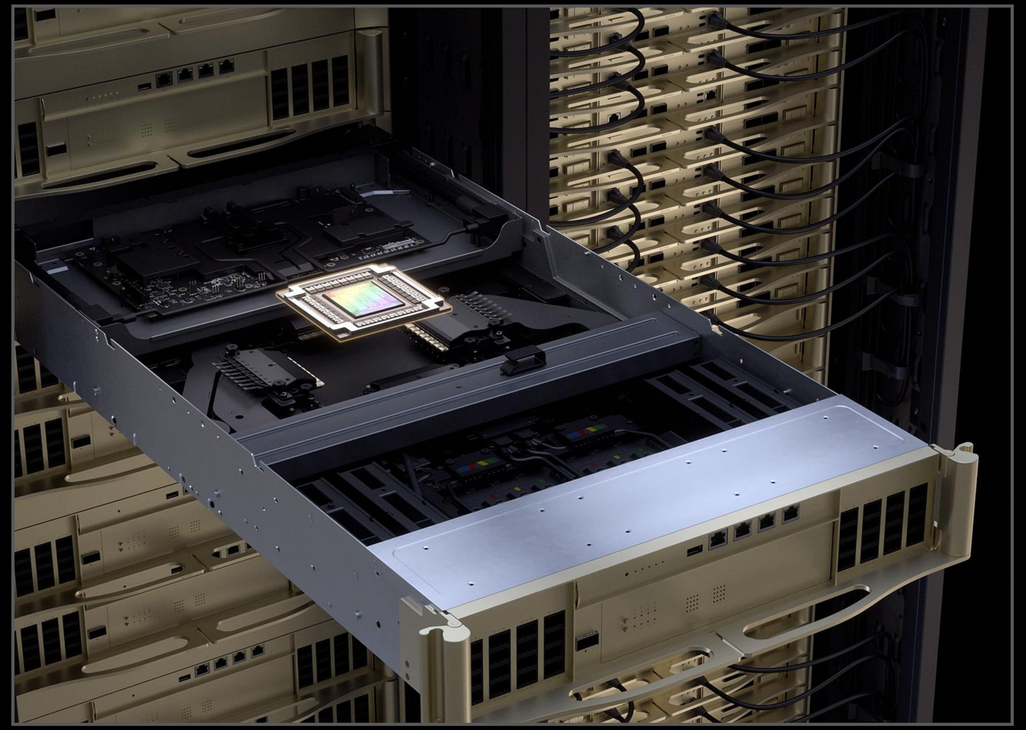
Scale-Up
NVLink



10X
Higher Packet Rate

3X
Lower Latency

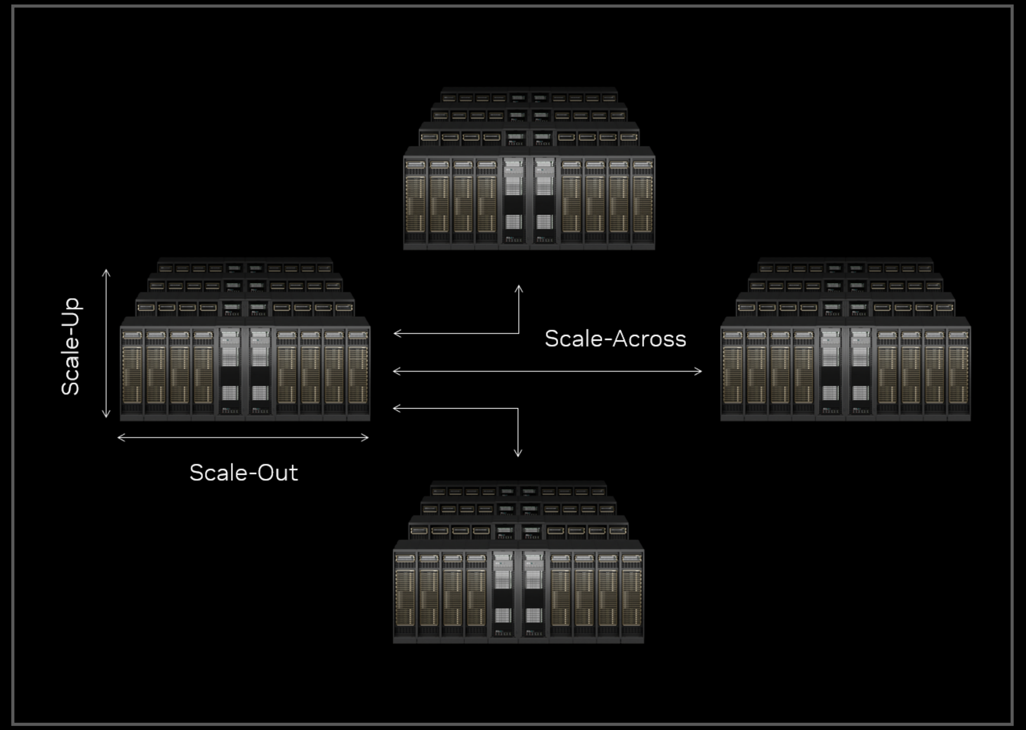
Scale-Out
Spectrum-X Ethernet



1.6X
Higher RDMA Bandwidth

3X
Lower Jitter

Scale-Across
Spectrum-XGS Ethernet



1.9X
Multi-Site Performance

Context Scale
Vera BlueField-4 Storage Processor

5X
Higher Token Throughput

5X
Greater Power Efficiency

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